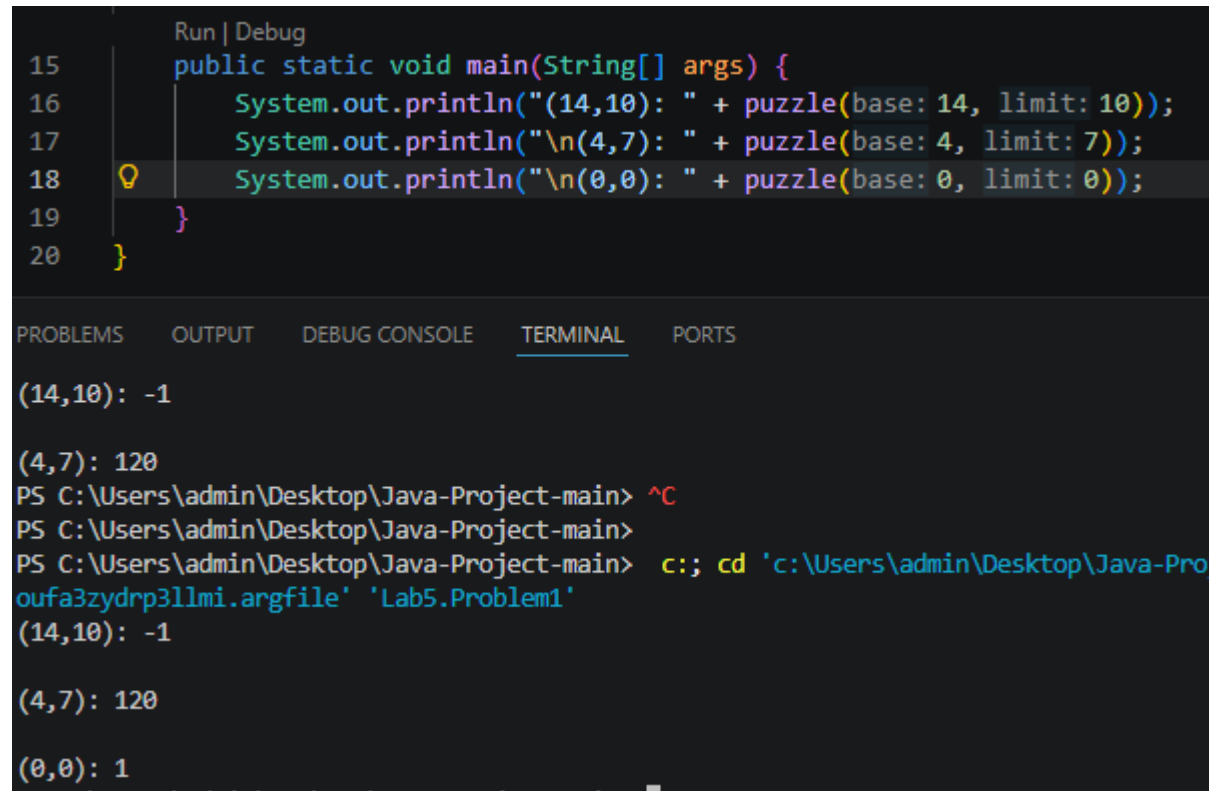


Problem 1

1/ The base case is the second if-else structure (base == limit)

2/ The function will be recursive when 'base < limit'

3/



The screenshot shows an IDE with a Java file. The code defines a `main` method that calls a `puzzle` function with three different sets of arguments: (14, 10), (4, 7), and (0, 0). The IDE's output window shows the results of these calls: (14,10) returns -1, (4,7) returns 120, and (0,0) returns 1. Below the output, a Windows command prompt is visible, showing the user navigating to the project directory and running the program.

```
Run | Debug
15 public static void main(String[] args) {
16     System.out.println("(14,10): " + puzzle(base: 14, limit: 10));
17     System.out.println("\n(4,7): " + puzzle(base: 4, limit: 7));
18     System.out.println("\n(0,0): " + puzzle(base: 0, limit: 0));
19 }
20 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(14,10): -1

(4,7): 120

PS C:\Users\admin\Desktop\Java-Project-main> ^C

PS C:\Users\admin\Desktop\Java-Project-main>

PS C:\Users\admin\Desktop\Java-Project-main> c:; cd 'c:\Users\admin\Desktop\Java-Pro
oufa3zydrp3llmi.argfile' 'Lab5.Problem1'

(14,10): -1

(4,7): 120

(0,0): 1

Problem 2

```
1  package Lab5;
2
3  public class Problem2 {
4      public static double sum(int n){
5          if (n <= 1) {
6              return 1;
7          }
8
9          return 1.0/n + sum(n-1);
10     }
11
12     Run | Debug
13     public static void main(String[] args) {
14         System.out.println(sum(n: 10));
15     }
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\admin\Desktop\Java-Project-main> ^C
PS C:\Users\admin\Desktop\Java-Project-main>
PS C:\Users\admin\Desktop\Java-Project-main> c:; cd 'c:
in32-x64\jre\21.0.10-win32-x86_64\bin\java.exe' '@C:\Use
2.9289682539682538
```

Problem 4

Findmin()

```
3 public class Problem4 {
4     public static int findmin(int a[], int n){
5         if (n == 1) {
6             return a[0];
7         }
8
9         int[] b = new int[n - 1];
10        for (int i = 1; i < n; i++) {
11            b[i - 1] = a[i];
12        }
13
14        return java.lang.Math.min(a[0], findmin(b, n - 1));
15    }
16
17    public static
    Run | Debug
18    public static void main(String[] args) {
19        // sample array
20        int a[] = {3,12,33,123,39,22,56,341,9,2};
21        System.out.println(findmin(a, a.length));
22    }
23 }
```

Terminal (Ctrl+')

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\admin\Desktop\Java-Project-main> c::; cd 'c:\Users\admin\Desktop\min\.vscode\extensions\redhat.java-1.54.0-win32-x64\jre\21.0.10-win32-x86Data\Local\Temp\cp_eaxqntir1foufa3zydrp3llmi.argfile' 'Lab5.Problem4'

2

Findsum()

```
16
17     public static int findsum(int[] a, int n){
18         if (n == 1) {
19             return a[0];
20         }
21
22         return a[n - 1] + findsum(a, n - 1);
23     }
    Run | Debug
24     public static void main(String[] args) {
25         // sample array
26         int a[] = {3,12,33,123,39,22,56,341,9,2};
27         // System.out.println(findmin(a, a.length));
28         System.out.println(findsum(a, a.length));
29     }
30 }
31
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\admin\Desktop\Java-Project-main> ^C
PS C:\Users\admin\Desktop\Java-Project-main>
PS C:\Users\admin\Desktop\Java-Project-main> c:; cd 'c:\Users\admin\Desktop\Java-Project-main\bin\java.exe' '@C:\Users\admin\AppData\Local\Microsoft\Windows\apps\640
```

Problem 6

```
1  package Lab5;
2
3  public class Problem6 {
4      public static int gcd(int p, int q){
5          return (q == 0) ? p : gcd(q, p % q);
6      }
7      Run | Debug
8      public static void main(String[] args) {
9          System.out.println(gcd(p: 6, q: 36));
10     }
11 }
```

ms (Ctrl+Shift+M)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\admin\Desktop\Java-Project-main> ^C
PS C:\Users\admin\Desktop\Java-Project-main>
PS C:\Users\admin\Desktop\Java-Project-main> c:: cd 'c:\
sions\redhat.java-1.54.0-win32-x64\jre\21.0.10-win32-x86
zydrp3llmi.argfile' 'Lab5.Problem6'
6
```

Problem 8

```

5 public class Problem8 {
6     public static void recursive_subset(ArrayList<Integer> a, int length){
7         for (int i = 0; i < length; i++) {
8             for (int j = 0; j < length; j++) {
9                 if (i >= j) {
10                     continue;
11                 }
12                 System.out.println(a.subList(i, j));
13             }
14         }
15     }
16
17     Run | Debug
18     public static void main(String[] args) {
19         // int[] raw_array = { 2, 4, 1, 6, 7, 2, 90, 23, 13, 8 };
20         int[] raw_array = {1,2,3,4,5};
21
22         ArrayList<Integer> a = new ArrayList<Integer>();
23         for (int element : raw_array) {
24             a.add(element);
25         }
26
27         recursive_subset(a, a.size());
28     }
29

```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

phatlinux@phatlinux:~/Desktop/Java-Project$ cd /home/phatlinux/Desktop/Java-Project ;
@/tmp/cp_7sp9z8iajl353hmmwrsiao0h7u.argfile Lab5.Problem8
[1]
[1, 2]
[1, 2, 3]
[1, 2, 3, 4]
[2]
[2, 3]
[2, 3, 4]
[3]
[3, 4]
[4]

```

Problem 10

```
Problem10.java X
src > main > java > Lab5 > J Problem10.java > Problem10 > displaySierpinski(int, double, double, double, double, double, double)

1 package Lab5;
2
3 public class Problem10 {
4
5     /**
6      * Recursively prints the points of the Sierpinski Triangle for a given order.
7      * @param order the depth of recursion
8      * @param p1 coordinate of first vertex
9      * @param p2 coordinate of second vertex
10     * @param p3 coordinate of third vertex
11     */
12     public static void displaySierpinski(int order, double x1, double y1, double x2, double y2, double x3, double y3) {
13         if (order == 0) {
14             // Base case: Draw the triangle
15             System.out.printf("polygon(%.2f, %.2f), (%.2f, %.2f), (%.2f, %.2f)%n", x1, y1, x2, y2, x3, y3);
16         } else {
17             // Recursive calls: Find midpoints
18             double x12 = (x1 + x2) / 2;
19             double y12 = (y1 + y2) / 2;
20             double x23 = (x2 + x3) / 2;
21             double y23 = (y2 + y3) / 2;
22             double x31 = (x3 + x1) / 2;
23             double y31 = (y3 + y1) / 2;
24
25             // Divide into three smaller triangles
26             displaySierpinski(order - 1, x1, y1, x12, y12, x31, y31);
27             displaySierpinski(order - 1, x12, y12, x2, y2, x23, y23);
28             displaySierpinski(order - 1, x31, y31, x23, y23, x3, y3);
29         }
30     }
31 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
phatlinux@phatlinux:~/Desktop/Java-Project$ cd /home/phatlinux/Desktop/Java-Project ; /usr/bin/env /usr/lib/jvm/java-21-openjdk-am
d64/bin/java @/tmp/cp_7sp9z8iajl353hmr5iao0h7u.argfile Lab5.Problem10
Sierpinski Triangle coordinates for order 3:
polygon(200.00, 20.00), (177.50, 65.00), (222.50, 65.00))
polygon(177.50, 65.00), (155.00, 110.00), (200.00, 110.00))
polygon(222.50, 65.00), (200.00, 110.00), (245.00, 110.00))
polygon(155.00, 110.00), (132.50, 155.00), (177.50, 155.00))
polygon(132.50, 155.00), (110.00, 200.00), (155.00, 200.00))
polygon(177.50, 155.00), (155.00, 200.00), (200.00, 200.00))
polygon(245.00, 110.00), (222.50, 155.00), (267.50, 155.00))
polygon(222.50, 155.00), (200.00, 200.00), (245.00, 200.00))
polygon(267.50, 155.00), (245.00, 200.00), (290.00, 200.00))
polygon(110.00, 200.00), (87.50, 245.00), (132.50, 245.00))
polygon(87.50, 245.00), (65.00, 290.00), (110.00, 290.00))
polygon(132.50, 245.00), (110.00, 290.00), (155.00, 290.00))
polygon(65.00, 290.00), (42.50, 335.00), (87.50, 335.00))
polygon(42.50, 335.00), (20.00, 380.00), (65.00, 380.00))
polygon(87.50, 335.00), (65.00, 380.00), (110.00, 380.00))
polygon(155.00, 290.00), (132.50, 335.00), (177.50, 335.00))
polygon(132.50, 335.00), (110.00, 380.00), (155.00, 380.00))
polygon(177.50, 335.00), (155.00, 380.00), (200.00, 380.00))
polygon(290.00, 200.00), (267.50, 245.00), (312.50, 245.00))
polygon(267.50, 245.00), (245.00, 290.00), (290.00, 290.00))
polygon(312.50, 245.00), (290.00, 290.00), (335.00, 290.00))
polygon(245.00, 290.00), (222.50, 335.00), (267.50, 335.00))
polygon(222.50, 335.00), (200.00, 380.00), (245.00, 380.00))
Ready
```

